



EDU TERIA

72nd BPSC

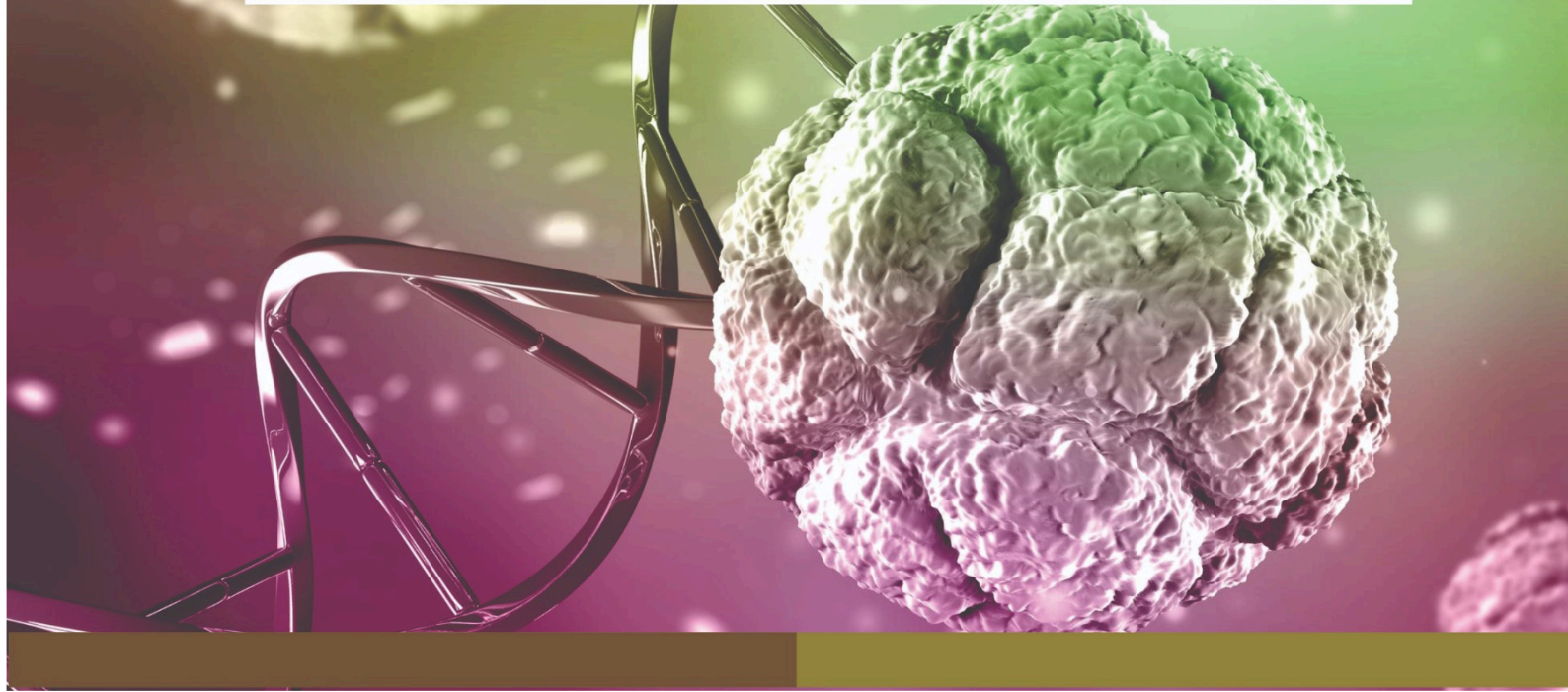
Prelims Online Batch

BIOLOGY

LECTURE

5

**Fertilization-Formation of fruits
and types of fruits**

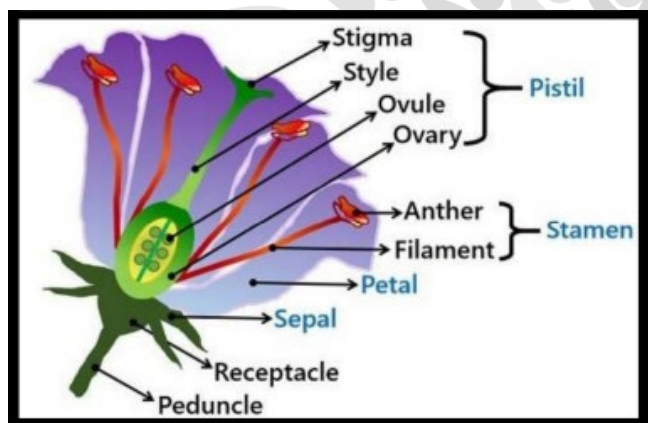


Fertilization-Formation of fruits and types of fruits

Fertilization in flower:

Generally flowering plants do sensual reproduction. sexual Reproduction = Process of reproduction in which gametes are involved and fertilization results into new species formation.

- Flower is the specialized reproductive structure of angiosperms, consisting of four main whorls: calyx (sepals), corolla (petals), androecium (stamens/male), and gynoecium (pistil/female).
- Key functions include attracting pollinators via colorful petals and producing seeds through fertilization.
- The ovary develops into fruit, while ovules become seeds.



- **Sepals (Calyx):** Outermost green, leaf-like structures protecting the flower bud.
- **Petals (Corolla):** Brightly colored parts attract insects and birds for pollination.
- **Stamen (Androecium - Male):** Produces pollen, consisting of an anther (pollen production) and a filament (stalk supporting the anther).
- **Pistil/Carpel (Gynoecium - Female):** Central part comprising : Stigma : Sticky top surface that receives pollen.
- **Style :** Tube connecting stigma to the ovary.
- **Ovary:** Basal part containing ovules, which develops into fruit/seeds after fertilization.

Flower: — Modified stem. Made up of four types of whorls:

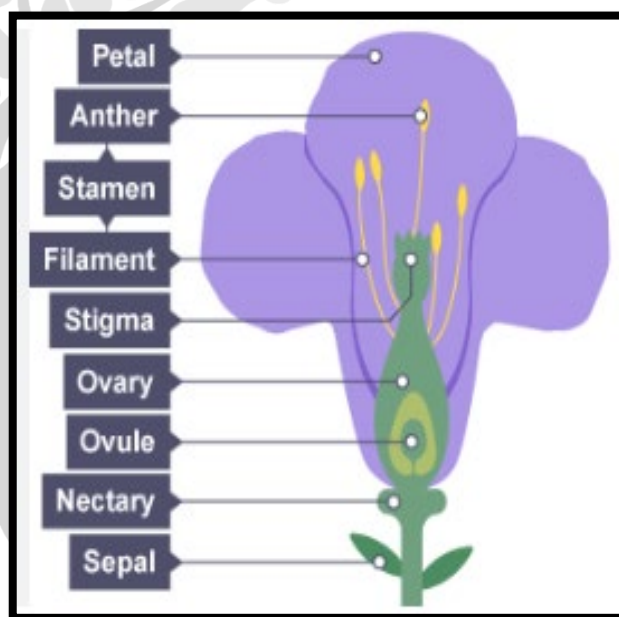
1. **Sepal (Group)**— Calyx
2. **Petal** — Corolla
3. **Stamen** — Androecium
4. **Carpal** — Gynoecium

Sexual reproduction in flowering plants includes following steps:

- I. Gamete formation
- II. Pollination
- III. Fertilization
- IV. Post fertilization

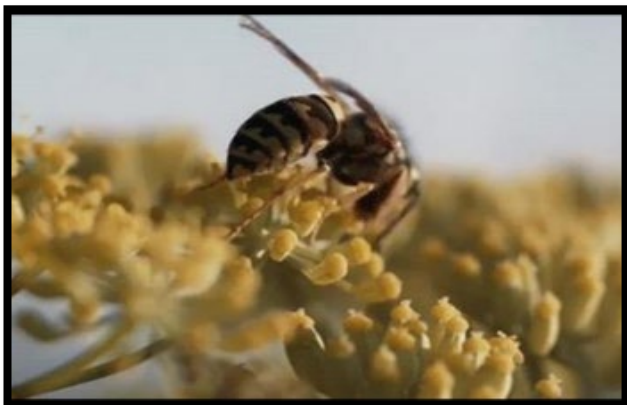
I. Gamete formation:

Formation of egg Cell in carpal/ Pistil. Formation of pollen grains in stamen.



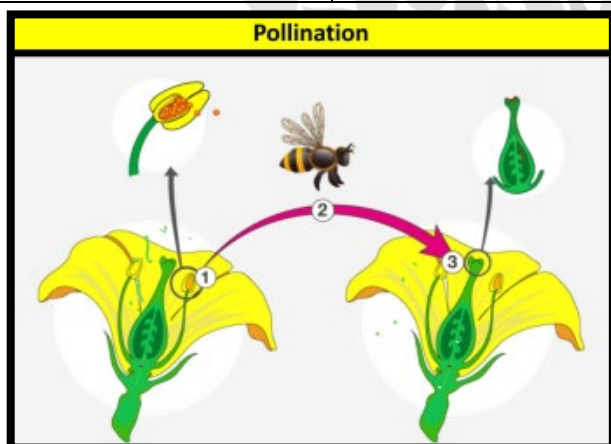
II. Pollination

- Pollination is the transfer of pollen grains from the anther to the stigma, essential for sexual reproduction in plants, categorized into self-pollination (Autogamy/Geitonogamy) and cross-pollination (Xenogamy).
- It requires external agents (wind, water, animals) and relies on compatibility between pollen and stigma (pollen-pistil interaction) to ensure fertilization.



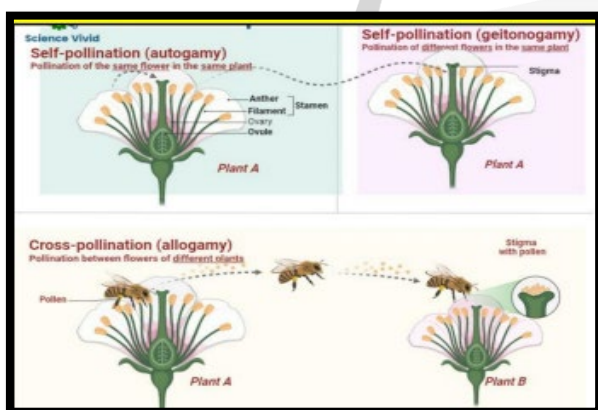
Pollination

Self Pollination	Cross Pollination (Xenogamy)
Transfer of pollen grain from anther to stigma with in the same plant	Transfer of pollen grain from anther to stigma between two plants of same species.



❑ **Autogamy (Self-pollination):** Transfer of pollen from the anther to the stigma of the same flower. It requires synchrony in pollen release and stigma receptivity.

❑ **Cleistogamous flowers:** Do not open at all (e.g. *Viola Oxalis Camelina*), ensuring self-pollination



- **Chasmogamous flowers:** Exposed anthers and stigma.
- **Geitonogamy:** Transfer of pollen from the anther to the stigma of another flower on the same plant.
- **Functionally crosspollination** (requires agents) but genetically self-pollination

III. Fertilization

When the pollen tube reaches the ovule, two male gametes are released.

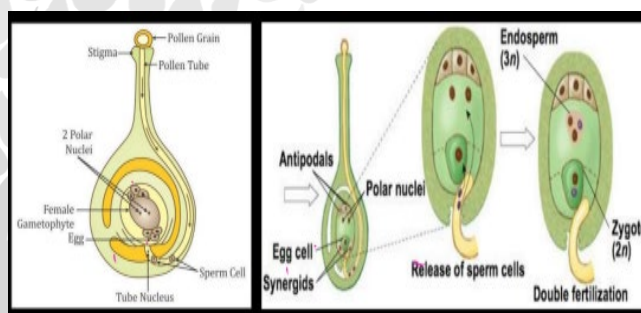
Flowering plants show Double fertilization:

1. Syngamy

Male gamete(n) + Egg Cell(n) → zygote (2n)

2. Triple fusion

Male gamete(n) + Polar Cell (n+n) → endosperm (3n)

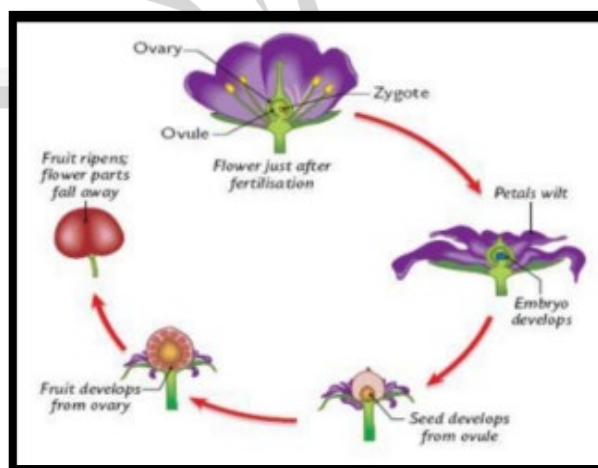


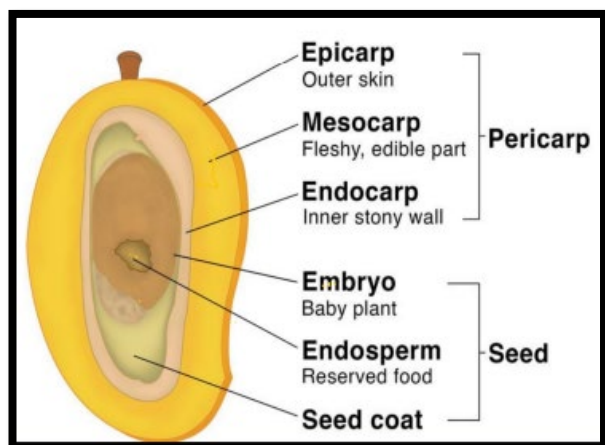
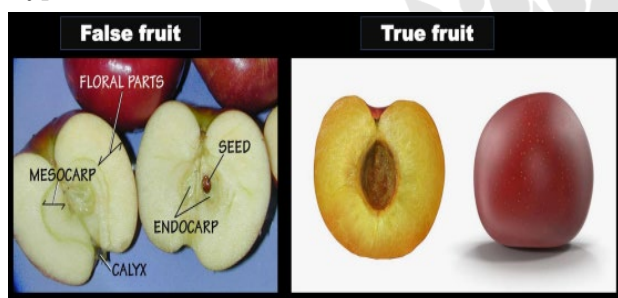
IV. Post Fertilization:

• **After fertilization: (post fertilization)**

- Ovary Fruit
- Ovule → Seed
- Zygote— Embryo

Formation of Fruit



**Parts of Fruit:****Types of Fruits**

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